9 Fig. 9.1 shows a wireless charging plate used to charge the battery in a mobile phone (cell phone). The coil of wire is part of an electric circuit.

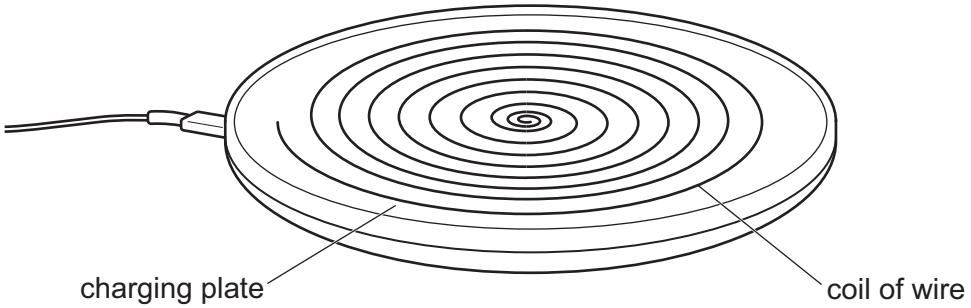

Fig. 9.1

The charging plate is connected to an a.c. power supply. The power supply is turned on.

(a) Describe the magnetic field around the charging plate in terms of its magnitude and direction.

.....

.....

..... [2]

(b) A mobile phone is placed on the charging plate as shown in Fig. 9.2. The coil in the mobile phone is part of a separate circuit that charges the battery.

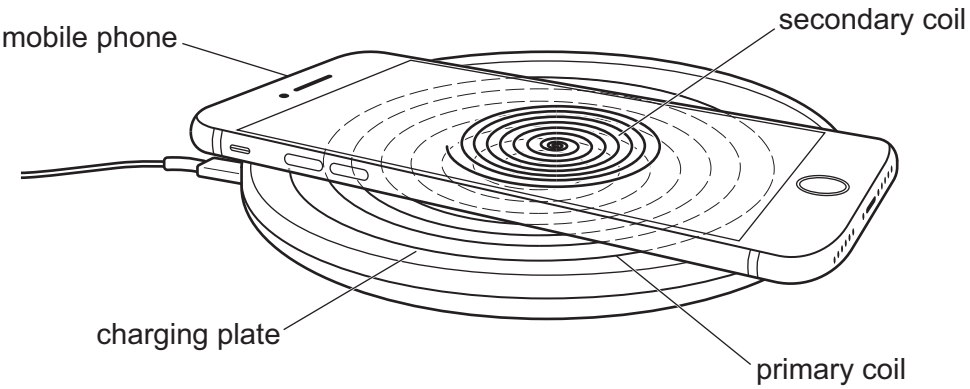

Fig. 9.2

The coil in the charging plate and the coil in the mobile phone act like a transformer.

(i) Explain why there is a current in the secondary coil shown in Fig. 9.2.

.....

.....

..... [2]

(ii) Suggest why the transformer made from the charging plate and mobile phone is not 100% efficient.

.....

..... [1]

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(c) The mobile phone battery can be recharged using this charging plate and stores 4.5×10^4 J of energy when fully recharged. The current in the secondary coil is 0.63 A when the output voltage is 12 V.

(i) Calculate the time taken to fully recharge a completely uncharged battery.

time = ... [2]

(ii) Calculate the charge passing through the battery in 60 s.

charge = ... [2]

[Total: 9]

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